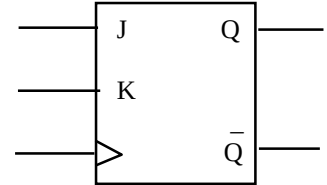


## Tutorial 6 Questions

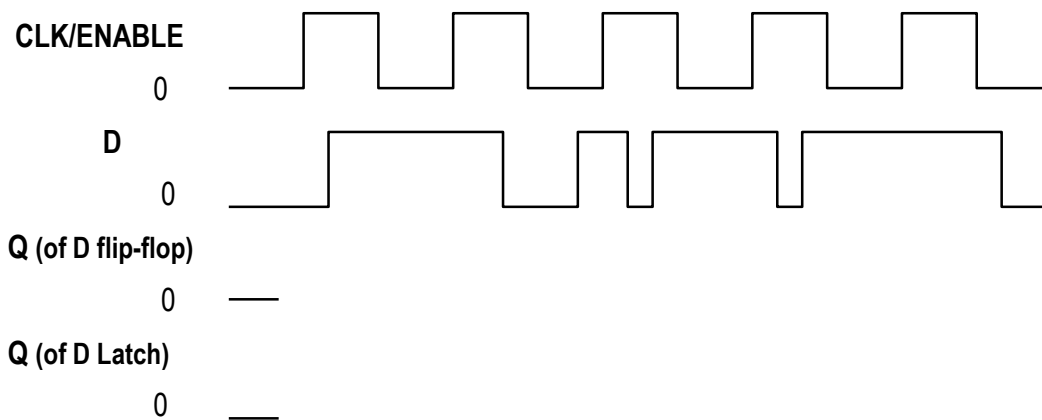
Please try out these self-study questions (labeled “SS”). These will not be discussed in class and solutions will be provided later.

### Flip Flops

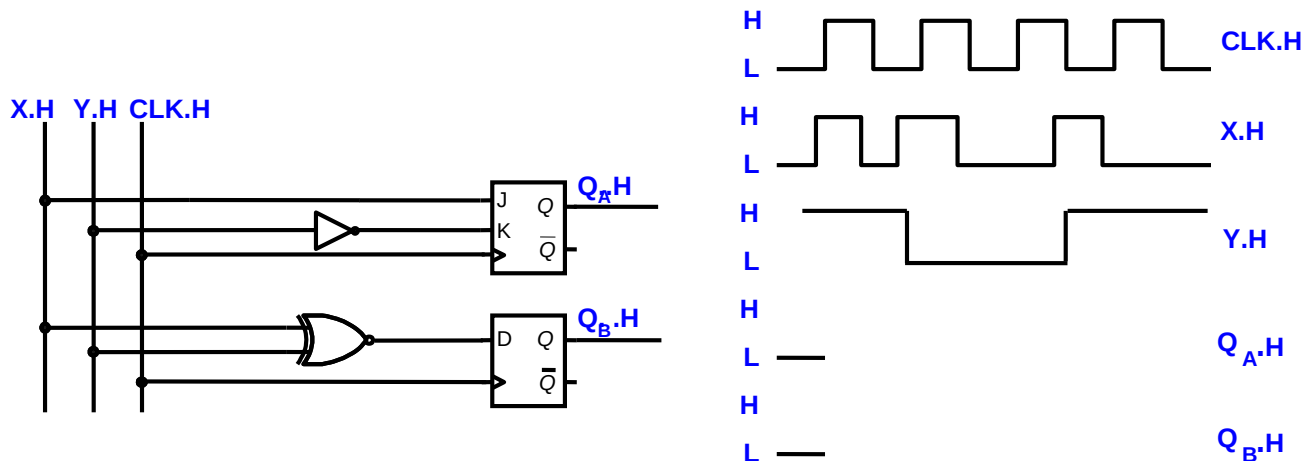
- SS1. Using additional logic gates, show how a JK flip-flop be modified to operate as a D flip-flop.
- SS2. Write a Verilog program to describe a JK flip-flop and check your solution using the Xilinx tools. (Refer to the characteristic table of the JK flip-flop for some help!)



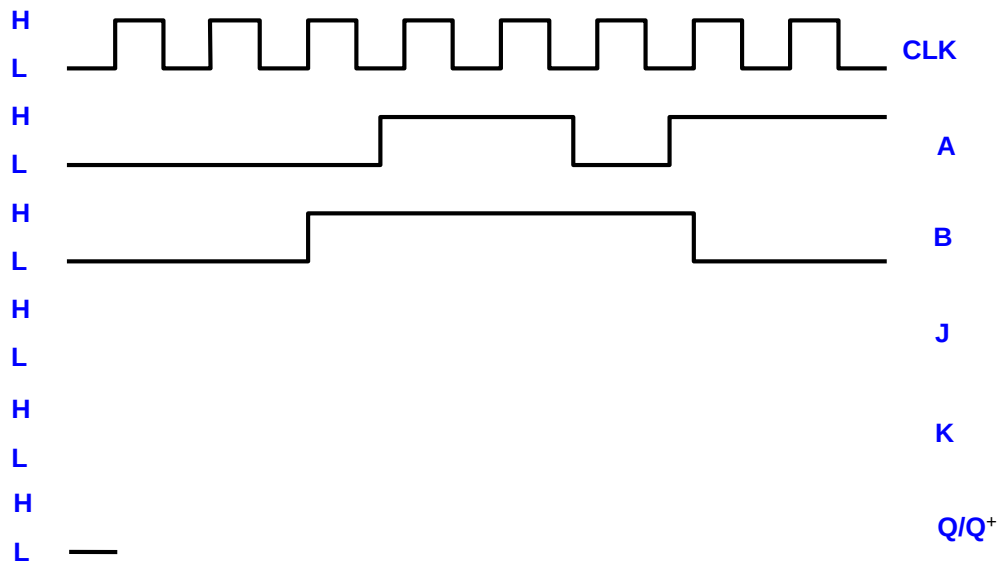
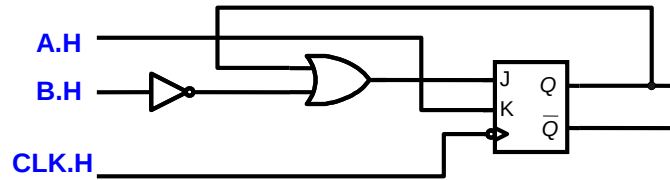
- Q1. The D and CLK/ENABLE input waveforms shown below are applied to a negative-edge triggered D flip-flop & a D Latch. Clearly draw the output Q waveforms of the devices and neglect all propagation delays.



- Q2. Given the circuit diagram below, complete the timing diagram below by filling in  $Q_A$  and  $Q_B$ . Include the propagation delays  $t_{PHL}$  and  $t_{PLH}$ .



- Q3. (a) Given the circuit diagram below, complete the timing diagram below by filling in J, K and Q. Neglect all propagation delays in this question.
- (b) The circuit below represents a type of AB flip flop, fill in its characteristic table and condensed characteristic table below.



| A | B | Q | Q <sup>+</sup> |
|---|---|---|----------------|
|   |   |   |                |
|   |   |   |                |
|   |   |   |                |
|   |   |   |                |
|   |   |   |                |
|   |   |   |                |
|   |   |   |                |
|   |   |   |                |

Characteristic Table

| A | B | Q <sup>+</sup> |
|---|---|----------------|
|   |   |                |
|   |   |                |
|   |   |                |
|   |   |                |

Condensed Characteristic Table